	S.S.V.P.S's B. S. Deore Polytechnic, Dhule **Department of Computer Engineering** *Academic Year : 2025 – 2026*

Presentation on Capstone Project Execution (CPE–316004)

Topic:	GoWheels- Online Multi-User Vehicle Rental & Listing Platform	
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Abstract

- GoWheels is a web-based vehicle rental and listing platform designed to simplify the process of renting vehicles such as cars and bikes.
- The system provides a digital platform where users can browse available vehicles, compare prices, check availability, and book vehicles online without visiting rental offices.
- This project aims to reduce manual work, eliminate booking conflicts, and improve transparency in vehicle rental services through automation and centralized data management.

Content

1. Introduction

2. Problem Statement

3. Scope of the project

4. Methodology

5. System Design & Working

6. Results and applications

7. Conclusion and future scope

8. References

1. Introduction

- Transportation is an essential part of daily life, but owning a vehicle is not always practical due to high costs and maintenance responsibilities. Traditional vehicle rental systems are often unorganized, relying on manual processes, physical paperwork, and limited availability information. This creates inconvenience for both customers and vehicle owners.
- The GoWheels platform addresses these issues by providing a centralized digital system where users can easily search, compare, and book vehicles. It also enables vehicle owners to list their vehicles and manage bookings efficiently.

2. Literature Review

- We focused on analyzing existing vehicle rental platforms such as Zoomcar, Revv, and Ola Drive, which provide digital booking and self-drive options. These platforms demonstrate the growing demand for online rental services and user convenience. However, most of them are limited to metro cities and do not effectively serve smaller towns.
- Additionally, they provide limited support for independent vehicle owners and often lack complete transparency in pricing and availability. A study of local rental agencies shows that they rely heavily on manual processes like registers, phone calls, and WhatsApp communication.

2. Literature Review

Market Metric	Data / Valuation
Estimated Market Size (2025)	₹2,60,586.8 Million (≈ ₹26,058.7 Crore)
Projected Market Size (2034)	₹8,32,481.7 Million (≈ ₹83,248.2 Crore)
Compound Annual Growth Rate (CAGR)	13.78% (2026–2034)
Top Revenue Segments	Economy Cars, Executive, & SUVs
Dominant Booking Mode	Online / App-based Bookings

2. Literature Review

Platform Name	Service Type	Key Features	Limitations
Zoomcar	Self-drive car rental	App-based booking, digital KYC, hourly/daily rentals	Limited availability in smaller cities
Revv	Self-drive & subscription	Doorstep delivery, flexible plans	Focused mainly on metro cities
Ola Drive	Self-drive rental	Integrated with Ola app, easy booking	Limited-service coverage
MyChoize	Self-drive & chauffeur	Corporate rentals, multiple vehicle options	Less penetration in tier-2 cities
Drivezy	Peer-to-peer rental	Vehicle sharing model, bikes & cars	Reduced operations in recent years
Savaari	Chauffeur-driven	Intercity travel, professional drivers	No self-drive option
Rapido Rentals	Bike rental	Affordable, quick booking	Limited to bikes and select cities

2. Literature Review

Problem Statement:

- Lack of real-time vehicle availability information
- Manual booking records and paperwork
- Difficulty in comparing vehicle rental options
- High chances of booking conflicts and errors
- Poor data management and record loss

Operational Task	Primary Method Used by Owners	Percentage of Owners
Inventory & Availability Tracking	Physical Registers / Diaries	68%
	Basic Excel Spreadsheets	24%
	Memorization / No Formal Tracking	8%
Customer Discovery & Booking	Word of Mouth / Walk-ins	45%
	WhatsApp Groups / Status	40%
	Justdial / Local Directories	15%
Document Verification (KYC)	Physical Photocopies (Stored in files)	85%
	Digital Photos on Phone Gallery	15%
Payment & Deposit Management	Cash Transactions	60%
	Direct UPI (Untracked ledger)	40%
Driver Allocation	Manual Phone Calls to Driver Network	100%

2. Literature Review

Problem Definition:

- To create a centralized system for listing and discovering vehicles easily.
- To implement digital KYC verification for secure user authentication.
- To provide a transparent booking system with clear pricing and availability.
- To develop an integrated wallet system for managing payments and transactions.
- To support both self-drive and driver-based rental options.
- To generate digital receipts and maintain booking history.
- To include a review and rating system for building trust between users.

3. Scope of the project

The scope of the GoWheels system includes the following functionalities:

- User registration and secure login system Browsing and filtering available vehicles based on category and availability
- Viewing detailed vehicle information including images and specifications
- Listing and managing vehicles by owners
- Selecting rental start and end dates for booking
- Automatic calculation of rental cost based on duration and options
- Uploading KYC documents (Aadhaar and Driving License) for verification
- Option to choose between self-drive and driver-based rentals
- Integrated wallet system for payments and transaction tracking
- Admin panel for managing users, vehicles, and bookings

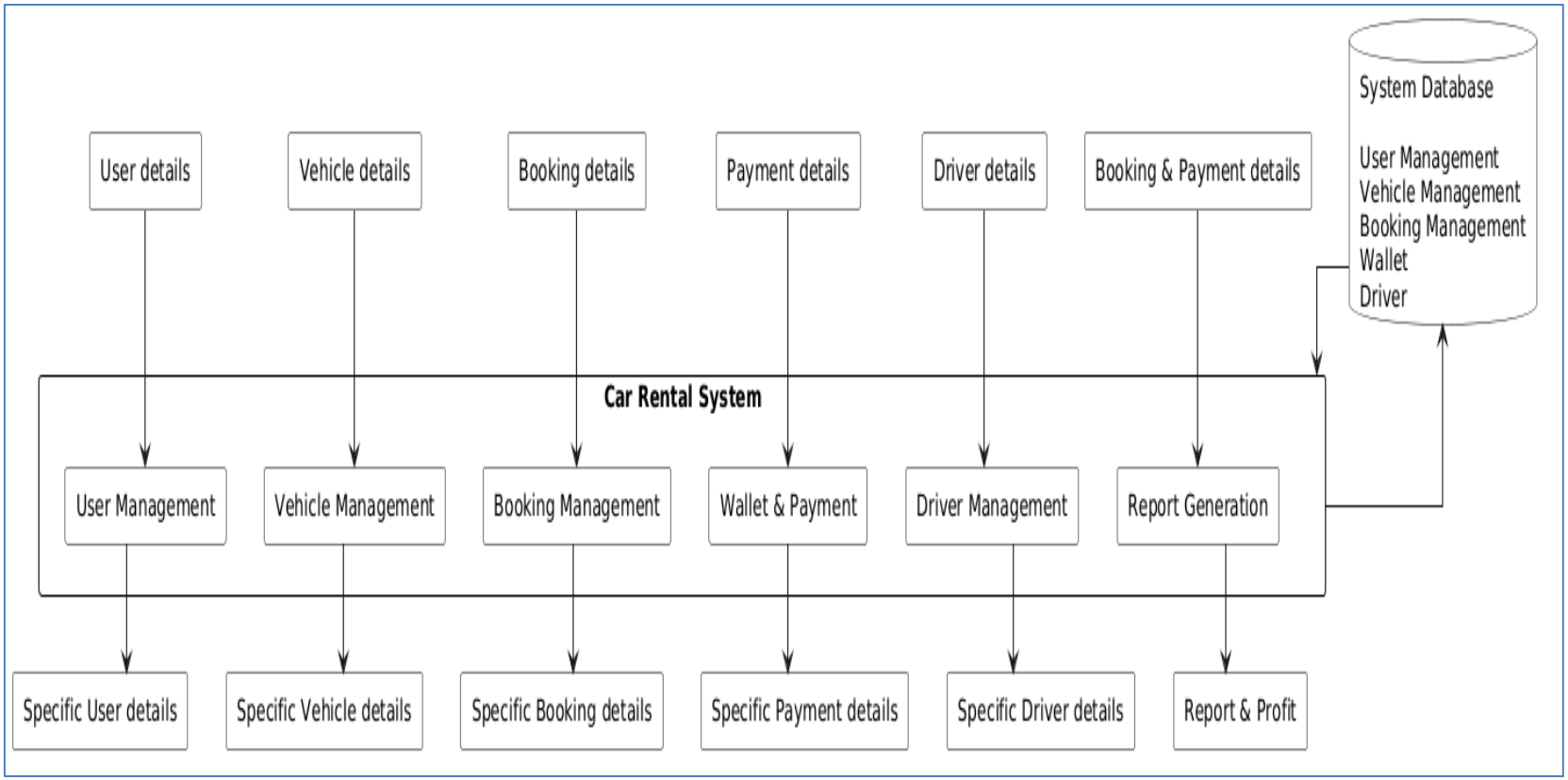
4. Methodology

Technologies used:-

- **Python:** Used as the core programming language for backend development.
- **Django Framework:** Used to build the web application and handle backend logic, database operations, and user authentication.
- **HTML, CSS, JavaScript:** Used to design and develop the frontend interface for better user interaction.
- **Bootstrap:** Used for creating a responsive and user-friendly UI design.
- **SQLite Database:** Used to store user data, vehicle details, bookings, and transactions.
- **Git & GitHub:** Used for version control and project management.
- **Selenium:** Used for testing the application and validating system functionality.

5. Details of designs, working & processes

- System diagram



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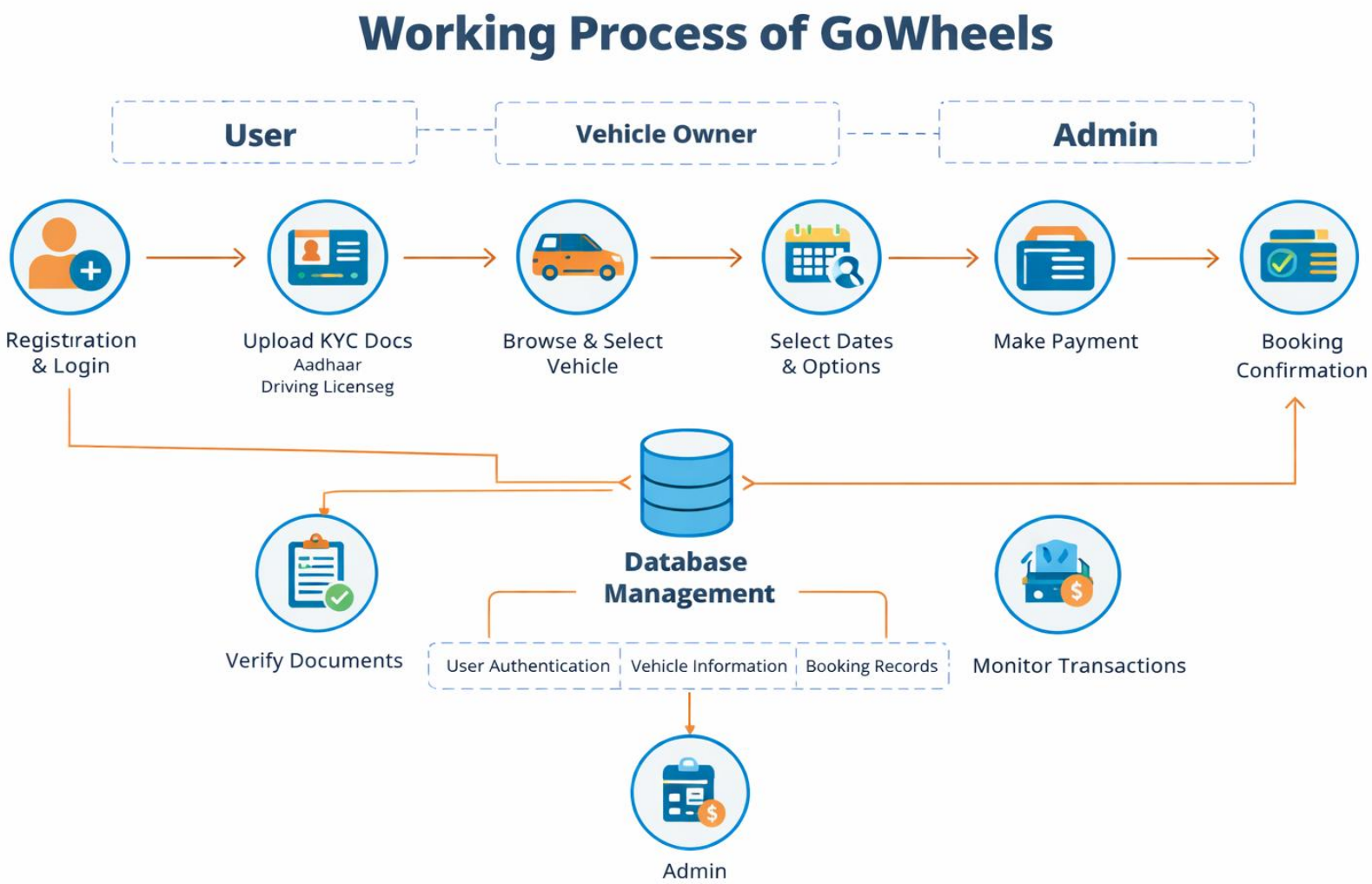

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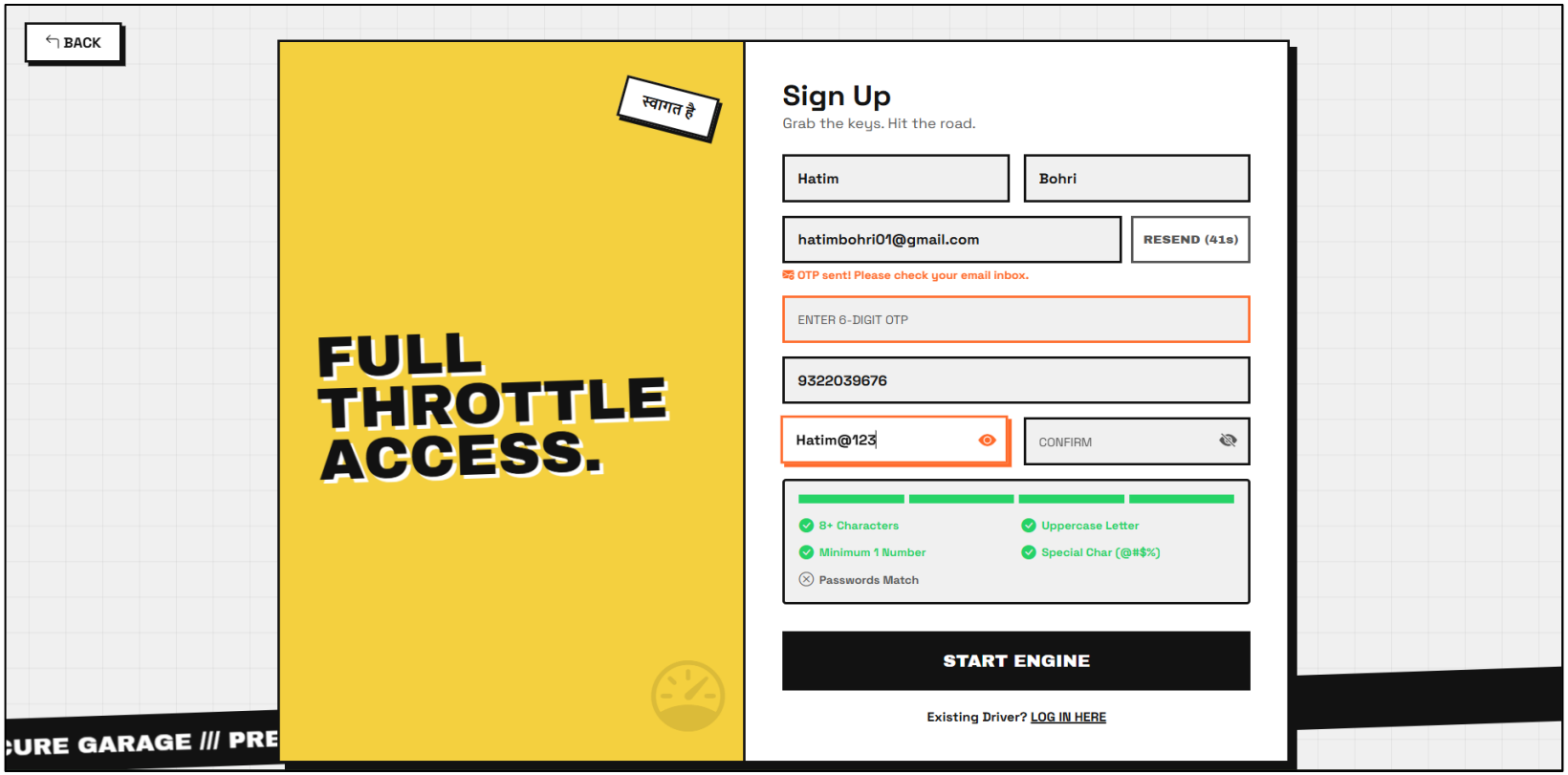

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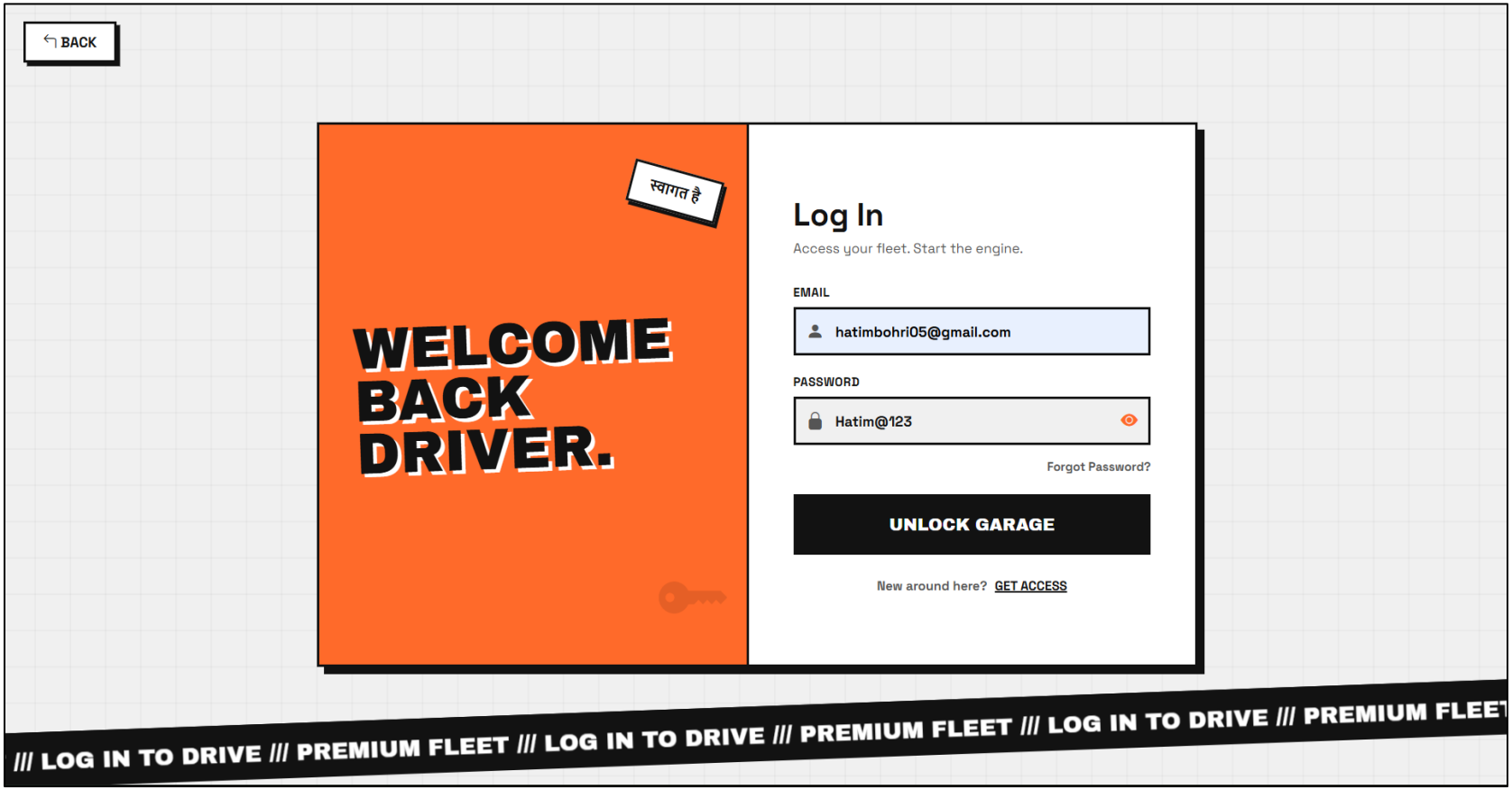

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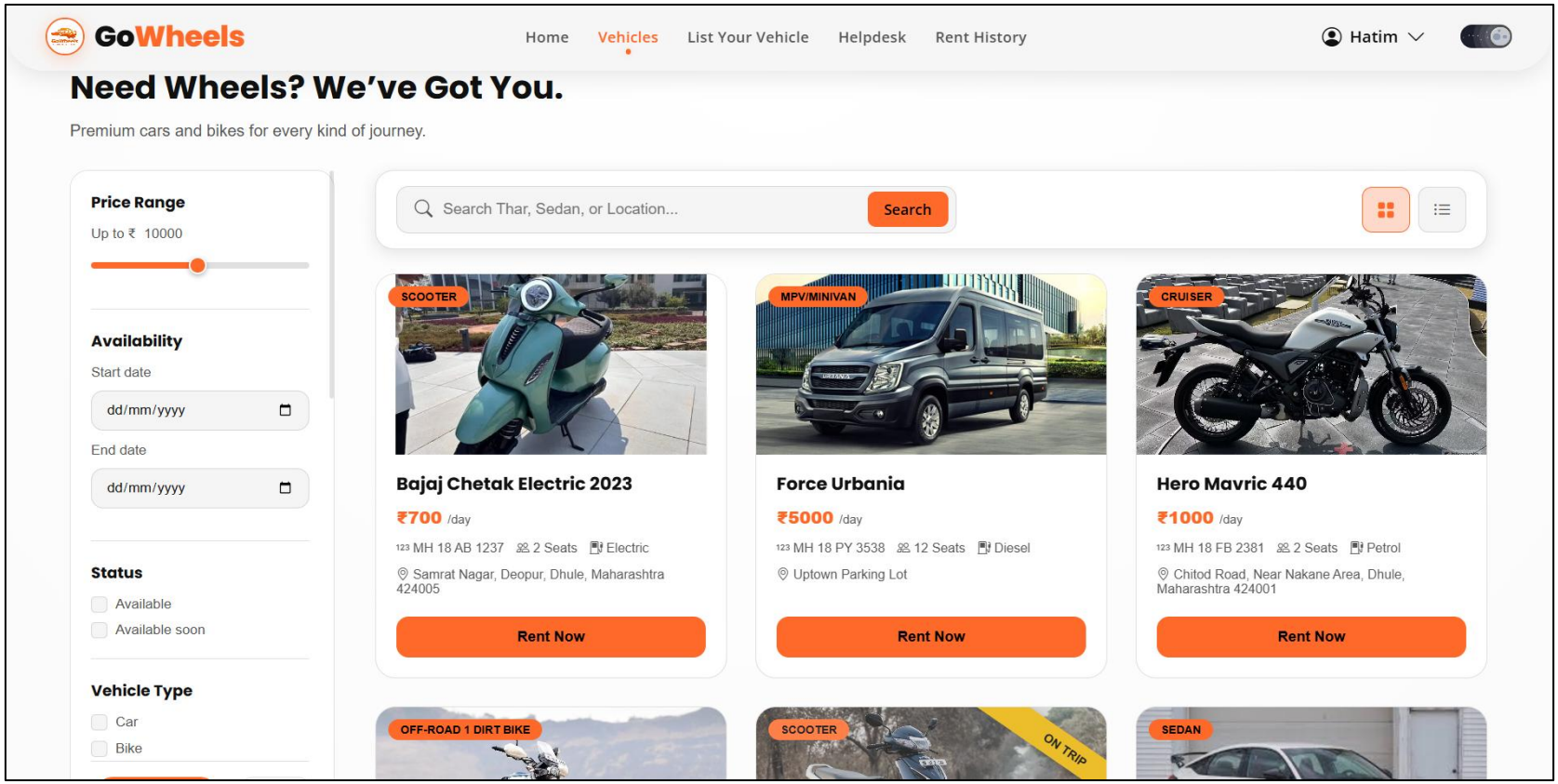

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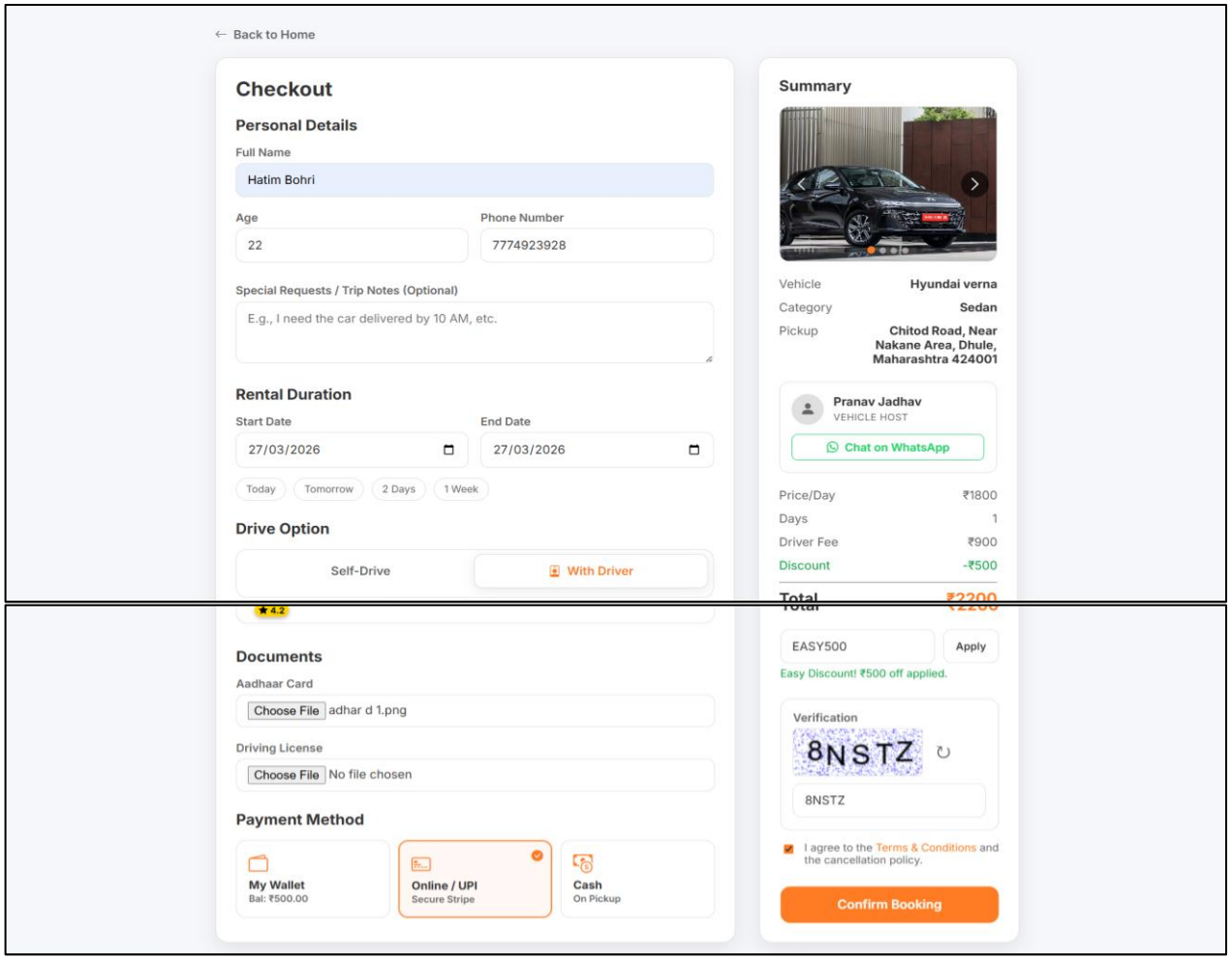

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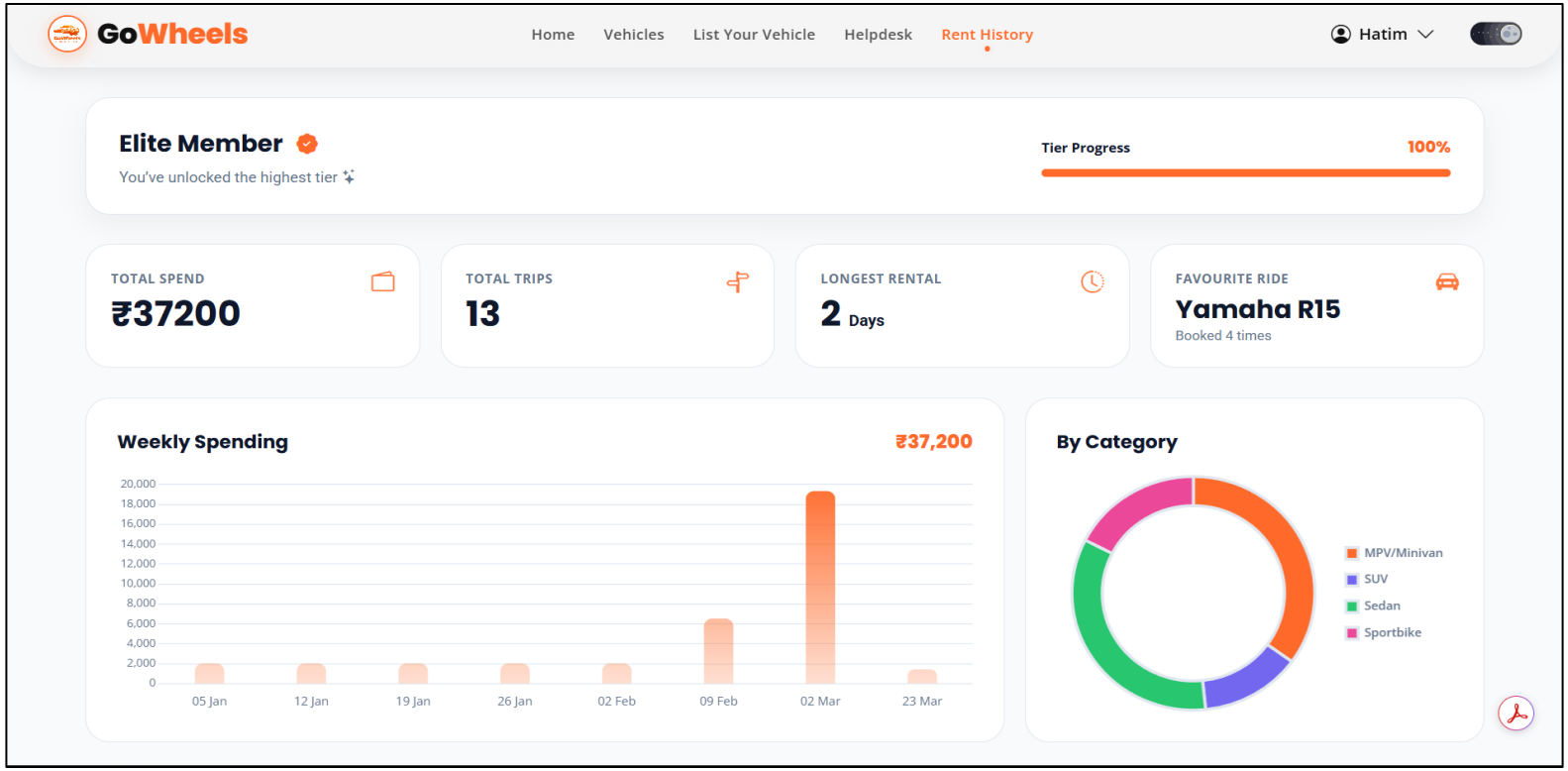

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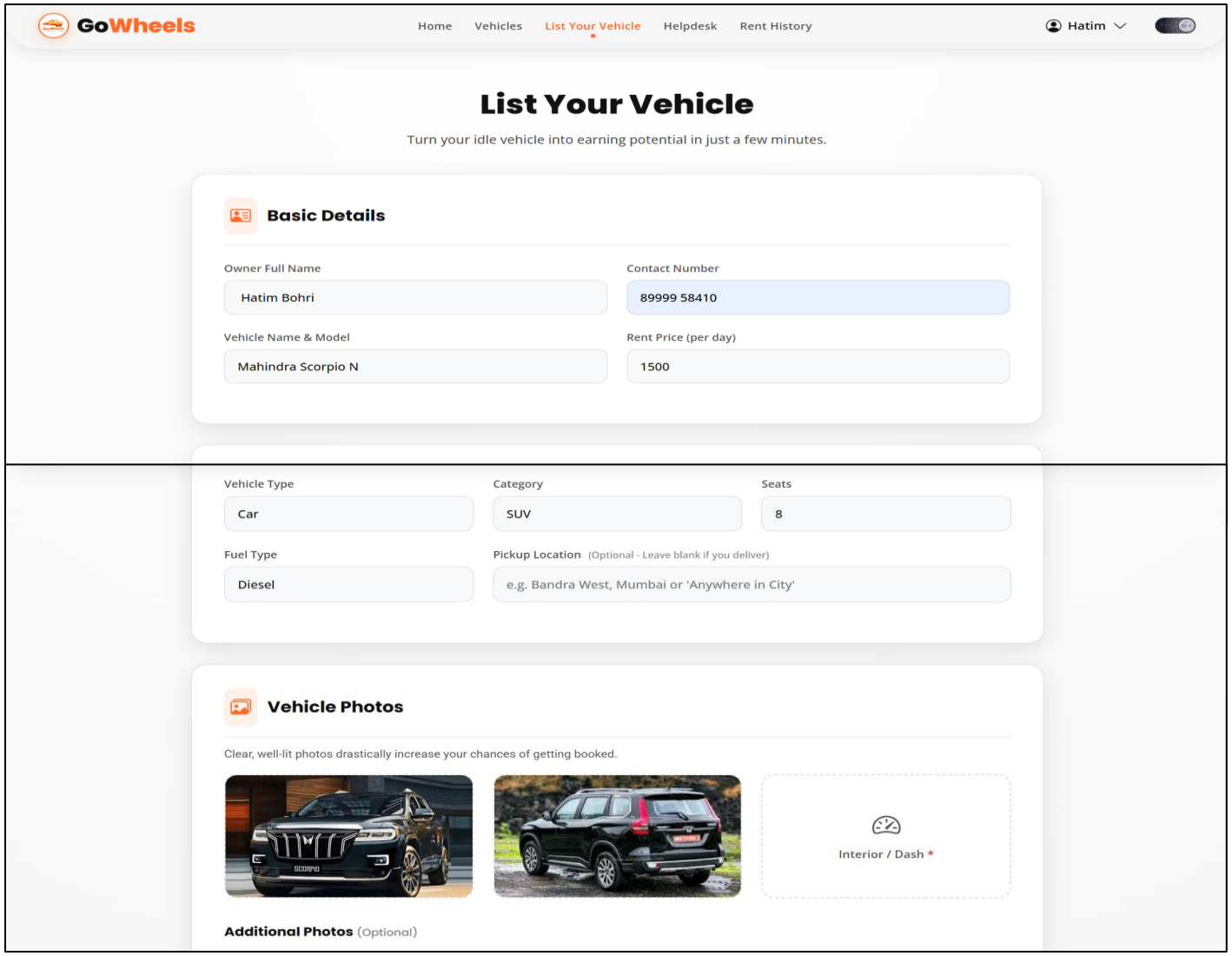

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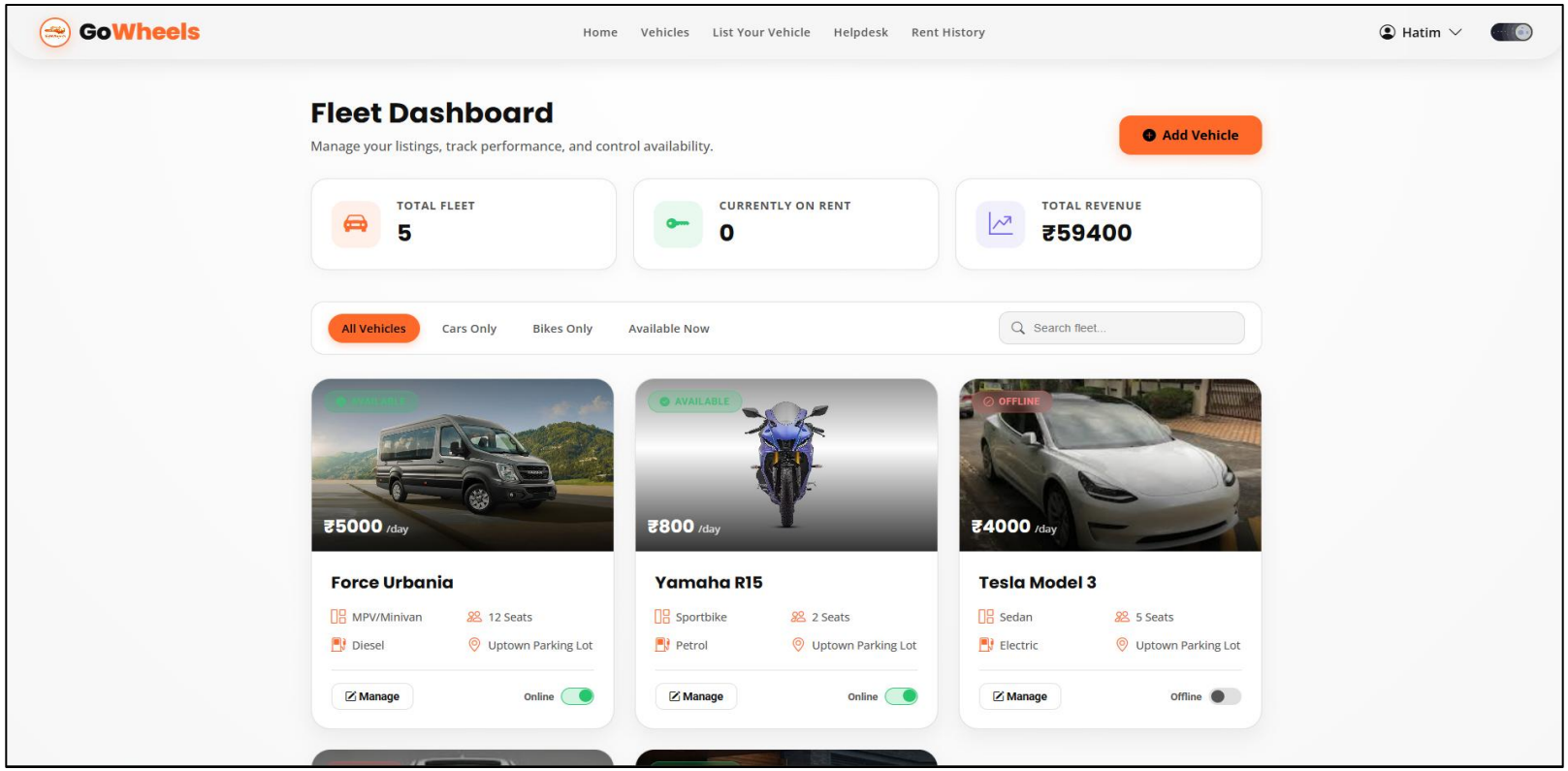
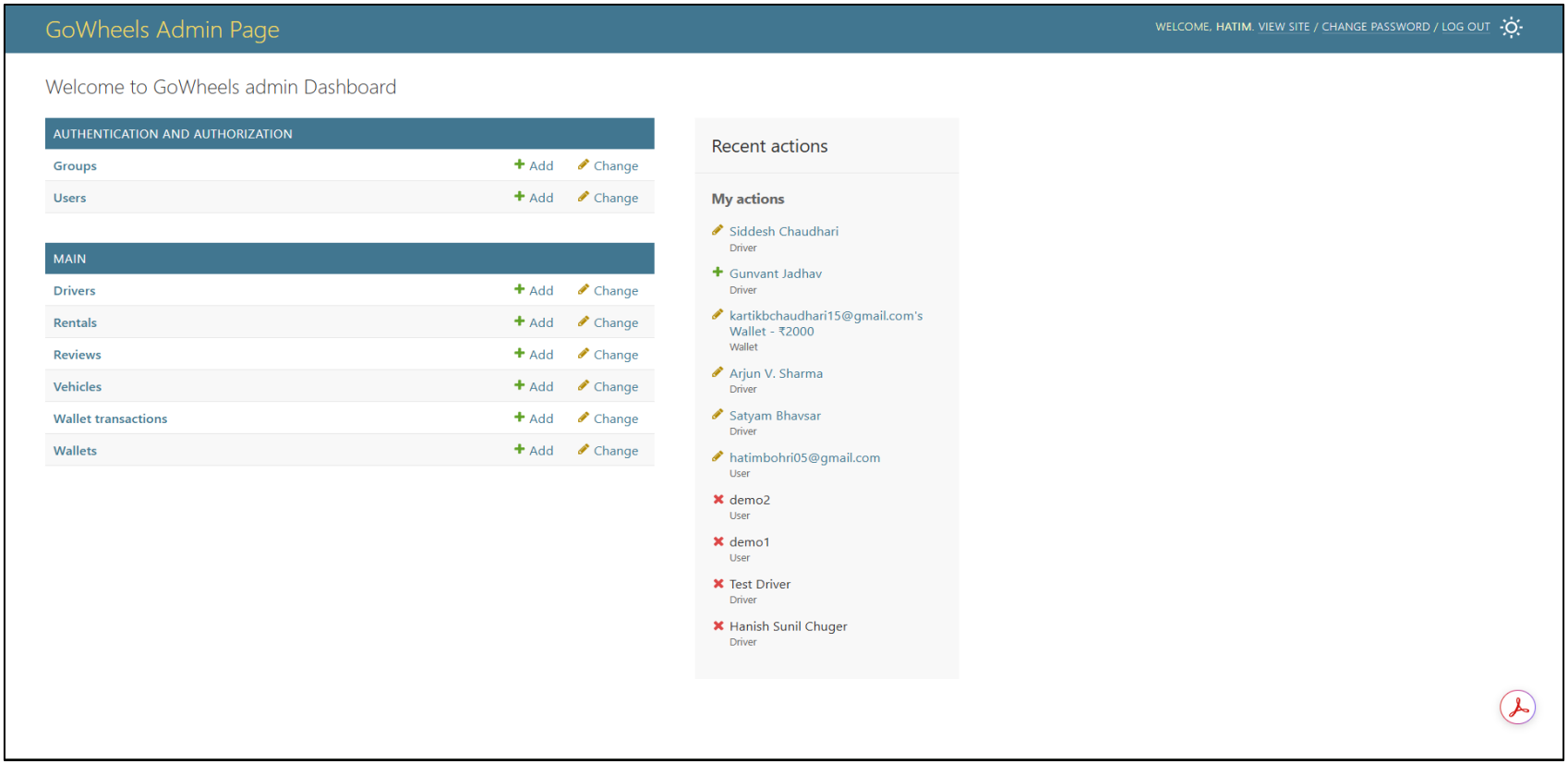


6. Results and Applications



7. Conclusion and future scope

Future Scope:-

- Development of a mobile application for Android and iOS platforms
- Integration of real-time GPS tracking for vehicle monitoring
- Implementation of AI-based dynamic pricing based on demand
- Expansion of the platform to multiple cities and regions
- Integration with secure online payment gateways (Razorpay, Stripe)
- Addition of multi-language support for wider accessibility
- Introduction of automated insurance and claim processing
- Advanced analytics for owners to track performance and revenue

7. Conclusion and future scope

- GoWheels successfully transforms the traditional vehicle rental system into a modern, digital platform. It simplifies booking, improves transparency, and ensures secure transactions between users and vehicle owners.
- The system not only enhances operational efficiency but also builds trust among users by providing reliable and structured services.

8. References and Bibliography

- **Django 4** By Antonio Melé
- **Python Crash Course** (A Hands-On, Project-Based Introduction to Programming) Eric Matthes
- **Official Django Documentation** (https://docs.djangoproject.com/)
- **Stripe API Official Documentation** (https://stripe.com/docs) - Studied for integrating the secure digital wallet and payment gateway.
- **Bootstrap 5 Official Documentation** (https://getbootstrap.com/) - Studied for designing the responsive, mobile-friendly frontend user interface.

- **Existing Vehicle Rental Applications Studied:**
- **Zoomcar** Developed by Zoomcar India Pvt. Ltd., representing the standard for self-drive car rentals in India.
- **Revv** Developed by PrimeMover Mobility Technologies Pvt. Ltd., studied for their flexible booking and delivery models.